

ACCTG 211H — Mechanics Drill Set (Answer Key)

Week 3 — Uncertainty and News

Week 3 drills: uncertainty and news

Use the stated risk-neutral model and a calculator. Complete after Day 5 and submit before Day 6; check and correct your work. Graded for completion; no generative AI.

1. A claim pays \$154 or \$88 in one year, each with probability 0.5. Calculate expected cash and value at 10%. Compare with a certain \$121 payment on the same date.
2. An all-equity enterprise distributes \$110 at the end of Year 1 and either \$154 or \$88 at the end of Year 2, with equal prior probabilities. The rate is 10%. The state is perfectly revealed at the end of Year 1. There is no terminal asset value, no debt, and no further investment. Calculate (a) initial value, (b) Year 1 closing asset and equity in each state, (c) Year 1 income in each state, (d) expected Year 1 income assessed at acquisition, and (e) the expected-return and news components of each outcome.
3. If Problem 2's state is revealed only in Year 2 and no other news arrives in Year 1, what are Year 1 closing value and income? Explain what moved to a later period.
4. At 5% for two years, compare a certain \$1,000 with a 90% chance of \$1,000 and a 10% chance of zero. How much value is lost? What change in the assumptions accounts for the difference in this calculation?
5. An unbiased forecasting process sometimes produces estimates above realized cash and sometimes below. Does one low realization establish manipulation? State one piece of evidence you would seek about the original estimate.

Self-Check Answers

1. Expected cash = **\$121**; value = **\$110**. The certain claim also has value \$110 under the supplied assumptions.
2. (a) $110/1.10 + 121/1.10^2 = \mathbf{\$200}$. (b) High **\$140**, low **\$80**, for both the remaining asset and equity; closing cash and liabilities are zero. (c) High income = $110 + 140 - 200 = \mathbf{\$50}$; low = $110 + 80 - 200 = \mathbf{-\$10}$. (d) **\$20**. (e) Expected return **\$20** in both states; news is **+\$30** or **-\$30**. First-year cash flows are the same in both states.
3. Closing value **\$110**, income **\$20**. With no Year 1 news, the state-dependent revision is postponed; ultimate operating cash flows are not changed.
4. Values **\$907.03** and **\$816.33**; difference **\$90.70**. Expected cash falls from \$1,000 to \$900. This is a discounted expected-payoff reduction, not an isolated risk premium or value of information.
5. **No**. Seek information available at the estimate date, model assumptions, evidence of bias, or consistent application. Do not judge solely with hindsight.